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NTU ADL 2025 Fall HW3

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Retrieval-Augmented Generation

CSIE5431 Applied Deep Learning

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1 Abstract

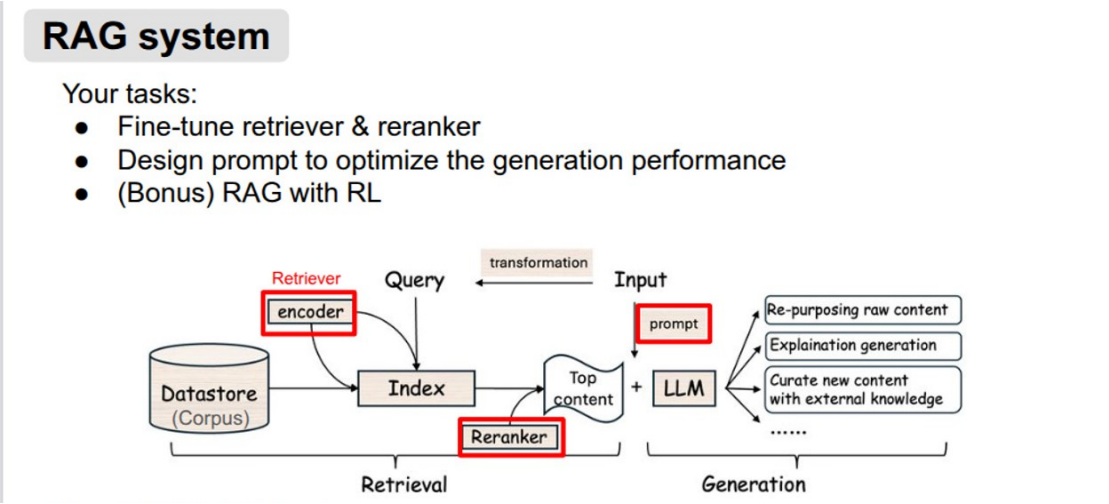


Figure 1: RAG system Architechture

2 Retriever Reranker Tuning

2.1 Training Data Construction

I constructed the training examples using train.txt, corpus.txt, and qrels.txt. Each query's rewrite field served as the anchor, while passages were obtained from the corpus. Only queries appearing in the training file were included.

2.2 Retriever

Base model: intfloat/multilingual-e5-small

2.3 Loss function

Retriever loss function

$$\mathscr{L}_{\text{ret}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\sum_{p\in P_i^+}\exp(s(q_i,p)/\tau)}{\sum_{p\in C}\exp(s(q_i,p)/\tau)}.$$

P_i^+: known positive passages in the candidate batch.
C: the batch's candidate passage pool.
$s(q_i,p)$: cosine similarity between normalized embeddings.
τ: temperature.

2.4 Reranker

Base model: ms-marco-MiniLM-L12-v2

2.5 Loss function

$$\mathscr{L}_{\text{rank}} = -\frac{1}{N}\sum_{j=1}^{N}\left[y_j \log \sigma(z_j) + (1-y_j)\log(1-\sigma(z_j))\right].$$

Here, z_j is the cross-encoder score and y_j is the relevance label. Explain that the model jointly processes each query and passage to distinguish relevant from irrelevant pairs.

3 Prompt Optimization

Table 1: Prompt variants and experimental objectives.

Experiment	Prompt change	What you investigate
P1	Minimal question-and-context instruction	Baseline behavior
P2	Add explicit short-answer and output-format constraints	Whether formatting improves answer similarity
P3	Require answers to rely only on the supplied context, with `CANNOTANSWER` when unsupported	Whether grounding reduces unsupported answers

4 Additional Analysis

Reranking improved MRR@10 from 0.6210 to 0.7454 across 3,153 held-out queries—an absolute gain of 0.1243, or 20.02% relative improvement

Table 2: Performance before and after reranking.

Setting	Recall@10	MRR@10
Before reranking	0.8925	0.6210
After reranking	0.8925	0.7454

5 Reference

1. Introduction to RAG.
2. E5 retriever and its training background.
3. Contrastive loss.